LaTeX Scribe Generation for Lecture Notes

Hi, you should behave as a professional scribe generator. Create a scribe as a LaTeX code of this file "L21_L22.pdf" file. preparing for my exam. It should be precise and accurate. Keep in mind that you should not change the content in the f the content accurate. Follow the lecture's structure and scope. Don't add the content from the reference file and also c (1).pdf" as the reference for the intended scribe's format. Make it possible so that it can be directly converted to LaTeX

Here is the LaTeX code for your scribe document based on the lecture slides. It is formatted to be precise, step-by-step, and ready fc

Code snippet

```
\documentclass{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath, amssymb}
\usepackage{geometry}
\geometry{a4paper, margin=1in}

\title{\textbf{School of Engineering and Applied Science (SEAS), Ahmedabad University} \\
\vspace{0.3cm}
\large CSE 400: Fundamentals of Probability in Computing \\
Lecture 21 \& 22 Scribe}
\author{Instructor: Dhaval Patel, PhD}
\date{March 24, 2026}

\begin{document}
\maketitle

\section{Chebyshev's Inequality 🚀}
\begin{itemize}
    \item \textbf{Core Concept:} It controls deviation from the mean by utilizing both mean and variance[cite: 13
    \item \textbf{Formula:} $P(|X-\mu|\ge a)\le\frac{Var(X)}{a^{2}}$[cite: 143].
    \item \textbf{Proof Steps:} Apply Markov's inequality directly on $(X-\mu)^{2}$[cite: 165].
    \item \textbf{Key Insight:} The bound decreases as the sample size $n$ increases, making it significantly tig
\end{itemize}

\section{Chernoff Bound 🔥}
\begin{itemize}
    \item \textbf{Core Concept:} Uses the exponential transformation $e^{tX}$ to amplify large tail values for a
    \item \textbf{Upper Tail Formula:} $P(X\ge a)\le \min_{t>0}\{\frac{E[e^{tX}]}{e^{ta}}\}$[cite: 504].
    \item \textbf{Lower Tail Formula:} $P(X\le a)\le \min_{t<0}\{e^{-ta}E[e^{tX}]\}$[cite: 713].
    \item \textbf{Steps to Solve:}
    \begin{enumerate}
        \item Transform the base inequality using $e^{tX}$[cite: 496].
        \item Apply Markov's inequality to the transformed variables[cite: 497].
        \item Optimize the inequality by choosing the best $t$ parameter to achieve the tightest bound[cite: 513]
    \end{enumerate}
\end{itemize}

\section{Jensen's Inequality 💡}
\begin{itemize}
    \item \textbf{Convexity Rule:} A function $g(x)$ is convex if $g(\alpha x_{1}+(1-\alpha)x_{2})\le\alpha g(x_{
    \item \textbf{Formula:} For a convex function $g(x)$ defined on an interval, $g(E[X])\le E[g(X)]$[cite: 863,
    \item \textbf{Interpretation:} Averaging the variables first results in a smaller value, while applying the c
\end{itemize}

\section{System Case Study: Server Crashing 💻}
\begin{itemize}
    \item \textbf{Problem Statement:} Sudden, extreme traffic spikes cause server queues to overflow, resulting i
    \item \textbf{Model Setup:} Define Demand as $X$ and Capacity as $C$. A system failure event occurs when $X \
    \item \textbf{Steps to Scale Capacity:}
    \begin{enumerate}
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        \item \textbf{Markov Bound:} If telemetry only provides the average load, the system requires a massive s
        \item \textbf{Chebyshev Bound:} Incorporating both average load and burstiness (variance) dramatically re
        \item \textbf{Chernoff Bound:} Knowing the full probability distribution enables precise server node sizi
    \end{enumerate}
    \item \textbf{The Golden Rule:} Adding more statistical information yields tighter bounds, which directly low
\end{itemize}

\end{document}
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